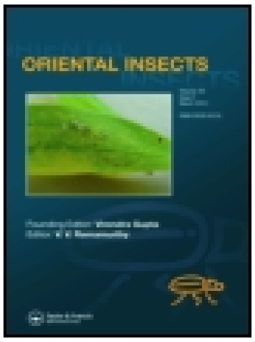

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Six species of Coccinellidae new to Bhutan (Insecta: Coleoptera)

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ABSTRACT

Six species of Coccinellidae representing two tribes: Coccinellini Latreille, 1807 [*Aiolocaria hexaspilota* (Hope, 1831), *Calvia championorum* Booth, 1997, *Calvia sykesii* (Crotch, 1874), *Halyzia nepalensis* Canepari, 2003, and *Megalocaria dilatata* (Fabricius, 1775)] and Noviini Mulsant, 1846 [*Novius octoguttatus* (Weise, 1910)] are reported for the first time from Bhutan. The male of *H. nepalensis* is described for the first time. Brief collection notes and pictorial illustration are provided.

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Introduction

The faunistic study of Bhutanese Coccinellidae has long been explored by naturalists around the world since as early as 1874 by Crotch. However, the knowledge on this well-studied group of beetles is still scant. In 2015, the first National Invertebrate Inventory has resulted in immense addition of invertebrate fauna found in the country, including Coccinellidae. There are currently 91 species of Coccinellidae reported from the country (Dorji et al. 2019). These are among large conspicuous species, while many smaller and problematic species are yet to be identified.

The Coccinellidae is broadly classified under two subfamilies: Microweiseinae Leng, 1920 and Coccinellinae Latreille, 1807. It was first proposed by Ślipiński (2007) in his book 'Australian Lady Beetles (Coleoptera: Coccinellidae)' and confirmed subsequently by phylogenetic analyses based on morphology and/or molecules (Seago et al. 2011; Robertson et al. 2015; Escalona et al. 2017). The most recent analysis of large molecular dataset indicates an existence of the third monotypic subfamily Monocoryninae (Che et al. 2021). The subfamily Coccinellinae includes most tribes and genera. In this paper, we present six conspicuous species representing two tribes:

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Coccinellini and Noviini. Some of these species are widely found in the region and identified relatively easily by their distinctive colour patterns. For one species, *Halyzia nepalensis*, the male is described for the first time. The species was first described by Canepari in 2003 from Nepal, based on a female type material at the Erfurt Natural History Museum, Germany. Recently, it has also been reported from Myanmar and India (Poorani and Thangjam 2019).

Materials and methods

The records presented are the result of a small collection effort during the year 2019. All the specimens were killed using ethyl acetate, relaxed for several hours and glued on cards after proper dissection of genitalia under a Nikon SMZ445 stereomicroscope. The genitalia were mounted next to the beetle on the same card.

Zeiss Axion Zoom V16 was used to take photographs of specimens and their genitalia. Photographs were edited using Adobe Photoshop software. The specimens studied are deposited in the College of Natural Resources (CNR), Royal University of Bhutan and the Naturalis Biodiversity Center, The Netherlands (RMNH).

Species accounts

The species are reported in alphabetic order. The original publications are given as well as any commonly used synonyms. For each record, the following information is presented: dzongkhag [=district], number and sex of specimens, registration code, locality, coordinates, altitude, collection date, collector biotope and collection. A brief diagnosis of the species and general distribution in the region are also provided along with short notes wherever applicable. Besides, to aid identification of the species, colour pictures of the habitus along with genitalia are presented.

Subfamily CoccinellinaeTribe Coccinellini
Aiolocaria hexaspilota (Hope 1831) (Fig. 1A-B).

Coccinella 6-spilota: Hope 1831: 30.
Aiolocaria hexaspilota: Crotch 1874: 178.

Material examined

Trashigang: 3f, CD205, Kanglung, Sherubtse, 27.28845°N 91.52404°E, 1833 m, 8.xi.2019, Merman Gurung, open field, (CNR).

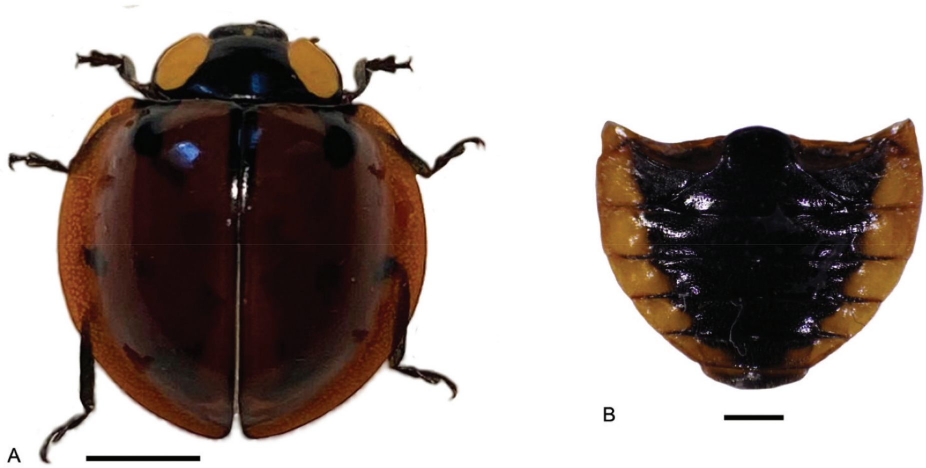

Figure 1A-B. *Aiolocaria hexaspilota* (Hope). A, habitus, dorsal view (scale bar–2 mm); B, abdomen with postcoxal lines (scale bar–1 mm).

Diagnosis

Members of the species large, with body length 10–14 mm, width 8–10 mm. Body convex dorsally and broadly oval. Pronotum black with two anterolateral yellowish spots. Elytra orange to brick red, with three black maculae; the first at longitudinal median line of each elytron and the second along suture and margin. Variants with different elytral spots and entirely black forms occurs (Crotch 1874).

Distribution

Nepal (Hope 1831), India, Pakistan, and Myanmar (Poorani 2002).

***Calvia championorum* Booth, 1997 (Fig. 2A-B).**

Calvia championorum: Booth 1997: 931.

Material examined

Punakha: 1f, CD123, Mitsina, Lobesa, 27.50359°N 89.87851°E, 1370 m, 14.iii.2019, C.Dorji, Chirpine forest, (CNR); Thimphu: 1f, CD207, Pangri Zampa, 27.53617°N 89.64043°E, 2490 m, 30.iii.2019, C.Dorji, Bluepine forest, (RMNH) .

Diagnosis

Medium size beetle with dull yellow to brownish-yellow ground colour. Weakly convex dorsally, margins of pronotum and elytra explanate. Specimens of this species sometimes have paler longitudinal stripes on

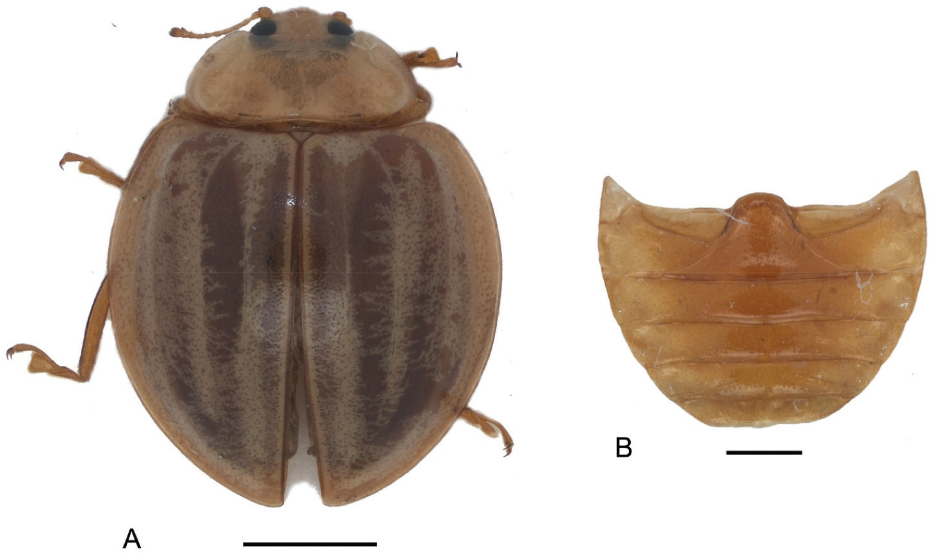

Figure 2A-B. Calvia championorum Booth. A, habitus, dorsal view (scale bar–2 mm); B, abdomen with postcoxal lines (scale bar–1 mm).

elytra; one adjacent to suture, one discal and one sublateral. Pronotum with M-shaped medial mark.

Comments

Eight of 13 species belonging to the genus *Calvia* Mulsant occurring in the Indian Subcontinent are now reported in Bhutan.

Distribution

India (Booth 1997).

Calvia sykesii **(Crotch 1874) (Fig. 3A-D).**

Anisocalvia sykesii: Crotch 1874: 146.
Calvia sykesi: Korschefsky 1932: 529.

Material examined

Punakha: 1m, CD122, Mitsina, Lobesa, 27.50359°N 89.87851°E, 1370 m, 25.ii.2019, C.Dorji, (CNR); Punakha: 1m, CD208, Mitsina, Lobesa, 27.51245°N 89.87055°E, 1381 m, 1.xii.2019, C.Dorji, Light trap, (CNR); Zhemgang: 1f, JW2, Buli, 27.13808°N 90.78017°E, 1628 m, 25.ii.2019, J. Wangchuk, Agricultural field, (CNR); Zhemgang: 1m, JW8, Tali, 27.17328°N 90.75436°E, 1759 m, 1.iii.2019, J. Wangchuk, Meadows, (CNR).

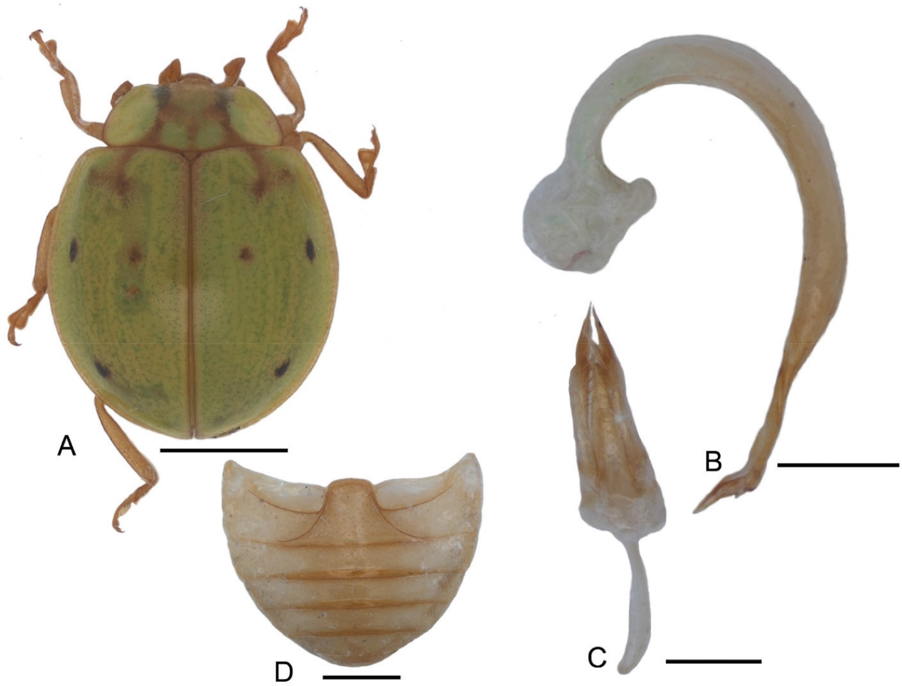

Figure 3A–D. *Calvia sykesii* (Crotch). A, habitus, dorsal view (scale bar–2 mm); B-C, male genitalia: B, penis, lateral view (scale bar–0.5 mm); C, tegmen, ventral view (scale bar–0.5 mm); D, abdomen with postcoxal lines (scale bar–1 mm).

Diagnosis

Length 6–8 mm, 4–5.5 mm broad. Body ovate, weakly convex dorsally. Generally greenish and tinged with red suture and lateral elytral margins. Elytron with three small distinct and irregular black marks: one near the margin at one-third, one nearer the suture in a transverse line with the first, one also near the margin at two-third. Pronotum with brownish red irregular but distinct M mark.

Comments

Apparently, the species was found attracted to light particularly during winter (November-February) at an altitude of about 1500 m.

Distribution

Nepal, India (Crotch 1874; Booth 1997).

***Halyzia nepalensis* Canepari, 2003** (Fig. 4A-E).

Halyzia nepalensis: Canepari 2003: 261.

Material examined

Punakha: 1m, CD120, Mitsina, Lobesa, 27.50359°N 89.87851°E, 1370 m, 19.xi.2018, C.Dorji, Light trap, (CNR); Punakha: 1m, CD208, Mitsina, Lobesa, 27.51245°N 89.87055°E, 1381 m, 1.xii.2019, C.Dorji, Light trap, (RMNH); Chukha: 1f, CD130, Darla, 26.86128°N 89.56669°E, 1737 m, 19.xi.2019, K. Wangdi, (CNR).

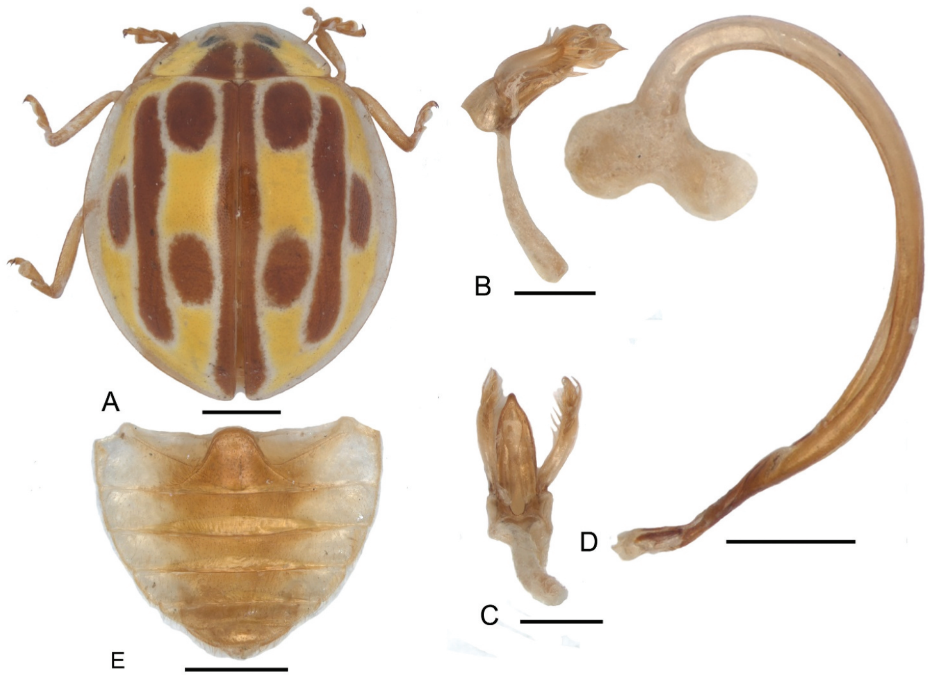

Figure 4A-E. Halyzia nepalensis Canepari. A, habitus, dorsal view (scale bar–1 mm); B-D, male genitalia: B, tegmen, lateral view (scale bar–0.5 mm); C, tegmen, ventral view (scale bar–0.5 mm); D, penis, lateral view (scale bar–0.5 mm); E, abdomen with postcoxal lines (scale bar–1 mm).

Diagnosis

General appearance of male is similar to female as described by Canepari (2003). Body oval, depressed. Pronotum and elytra pale-yellow with several brown markings; a pair of basilar spots on pronotum, a sutural border and four spots on each elytron. Apparently, sibling species like *H. straminea* share these features and at first glance, one may misidentify the species.

Male genitalia: Tegmen with penis guide symmetrical, not pointed apically; parameres broad and curved inward, slightly shorter than penis guide, separated, articulated with phallobase, with long setae on the apical part of margin; tegminal strut long rod-like, slightly curved inward. Penis base with distinct T-shaped capsule; penis uniformly curved with posterior inner margin enlarged; penis apex divided with conspicuous translucent tissue (Fig. 4D).

Distribution

Nepal (Canepari 2003), India and Myanmar (Poorani and Thangjam 2019).

***Megalocaria dilatata* (Fabricius 1775) (Fig. 5A-D).**

Coccinella dilatata: Fabricius 1775: 82.
Anisolemnia dilatata: Korschefsky 1932: 269.
Megalocaria dilatata: Iablokoff-Khnzorian 1982

Material examined

Punakha: 1f, CD195, CNR Academic Block, 27.50401°N 89.87830°E, 1376 m, 19. viii.2019, C.Dorji, Chirpine forest, (CNR).

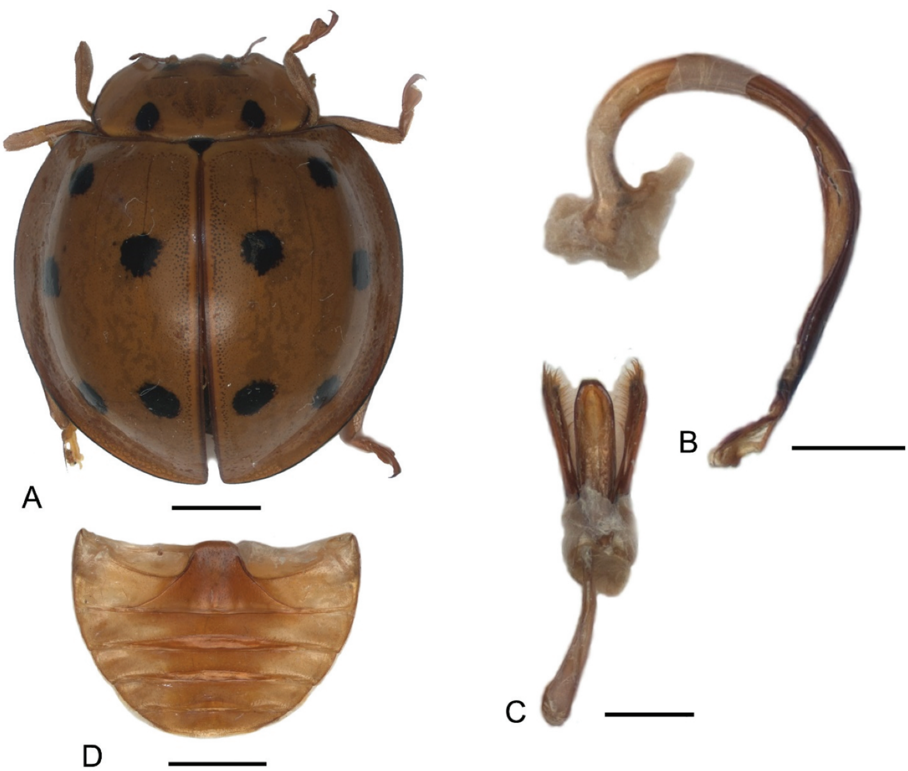

Figure 5A–D. Megalocaria dilatata (Fabricius). A, habitus, dorsal view (scale bar–2 mm); B-C, male genitalia; B, penis, lateral view (scale bar–1 mm); C, tegmen, ventral view (scale bar–1 mm); D, abdomen with postcoxal lines (scale bar–2 mm).

Diagnosis

Large and brightly orange coloured. Dorsally convex, hemispherical in outline. Species is known by 10 unequal, conspicuous circular spots arranged in 1-2-2 on each elytron. Pronotum with 2 latero-basal black spots.

Distribution

India (Poorani 2002). Nepal (Canepari 1997).

Tribe Noviini

Novius octoguttatus (Weise 1910) (Fig. 6A-D).

Rodolia 8-guttata: Weise 1910: 51.
Novius octoguttatus: Pang et al. 2020: 20.

Material examined

Thimphu: 1m, CD88, Buddha Point, 27.44453°N 89.64680°E, 2535 m, 8. vii.2018, C.Dorji, Chirpine forest, (CNR).

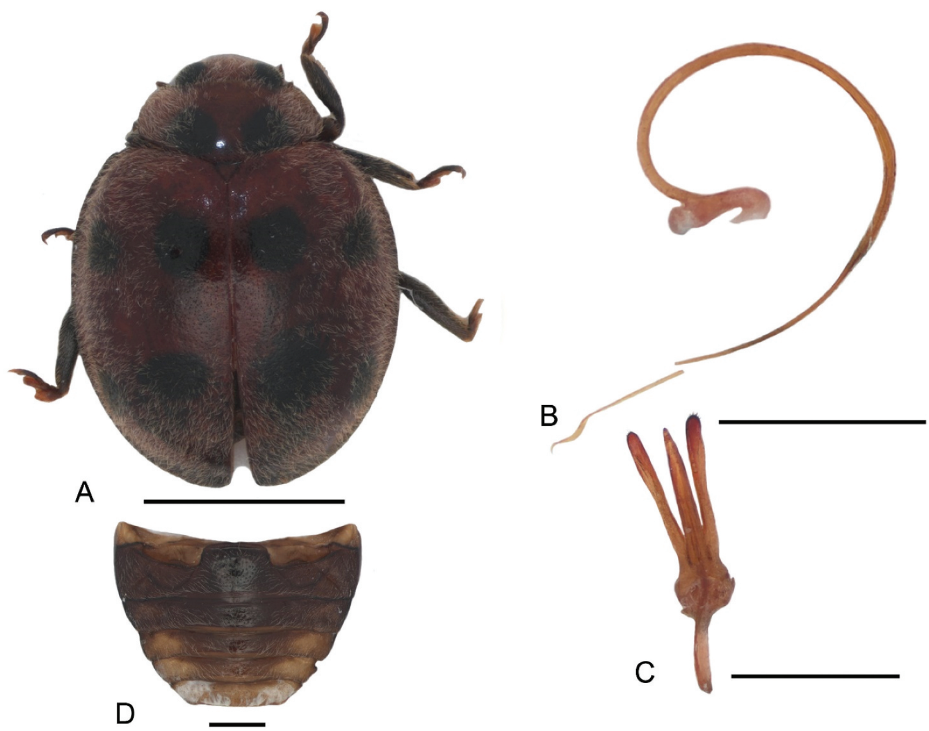

Figure 6A–D. Novius octoguttatus (Weise). A, habitus, dorsal view (scale bar–2 mm); B-C, male genitalia; B, penis, lateral view (scale bar–1 mm); C, tegmen, ventral view (scale bar–1 mm); D, abdomen with postcoxal lines (scale bar–0.5 mm).

Diagnosis

Medium size beetles, sub-hemispherical; dorsum pubescent. A pair of large black spots near the base of pronotum on either side of median line. Each elytron with 4 black spots; the first near scuteller angle, second below the shoulder, the third near the suture immediately below the transverse median line and the fourth subapical. Male genitalia: similar to *N. fumidus* (Mulsant).

Comments

The genus *Rodolia* was recently synonymised with *Novius* (Pang et al. 2020). This is the only species representing the genus in Bhutan.

Distribution

Nepal (Bielawski 1972), India (Kapur 1951), Myanmar (Poorani 2002), Pakistan (Iqbal et al. 2018).

Conclusions

Six additional species of lady beetles belonging to the subfamily Coccinellinae namely, *Aiolocaria hexaspilota, Calvia championorum, Calvia sykesii, Halyzia nepalensis, Megalocaria dilatata* and *Novius octoguttatus*, are reported from Bhutan. This brings the total number of lady beetles known in the country to 97 species. The coccinellid diversity recorded in Bhutan is relatively lower compared to neighbouring Himalayan countries owing particularly to limited research exploration. The international conservationists do not have easy access to research permits due to geopolitical and other reasons whilst the national researchers have limited capacity and resources. Promoting interest in research and building capacity of the national researchers through international collaboration and government support are critical in advancing the flora and fauna identification initiatives in the country.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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